

FINAL HOMEWORKS #7 & #8

HW #7 – Excel based- posted

(Portfolio Optimization & Bond Duration & Beta)

HW #8 – Calculus Review for the Final

(Pearson – now posted on Canvas)

BOTH will be due Thursday May 1st

FINAL Exam – Monday 5/5 (KOBL 100)***

Diversification

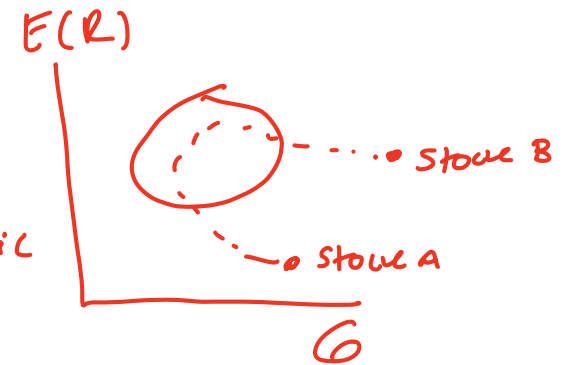
Systematic and Unsystematic Risk

6
Total risk has two components:

* systematic risk (market) *

unsystematic risk (firm specific risk)

we
measure
through
Beta



Reduction of unsystematic risk can be achieved through diversification

measured = total risk

Two Types of Risk

Systematic Risk (Market Related Risk)

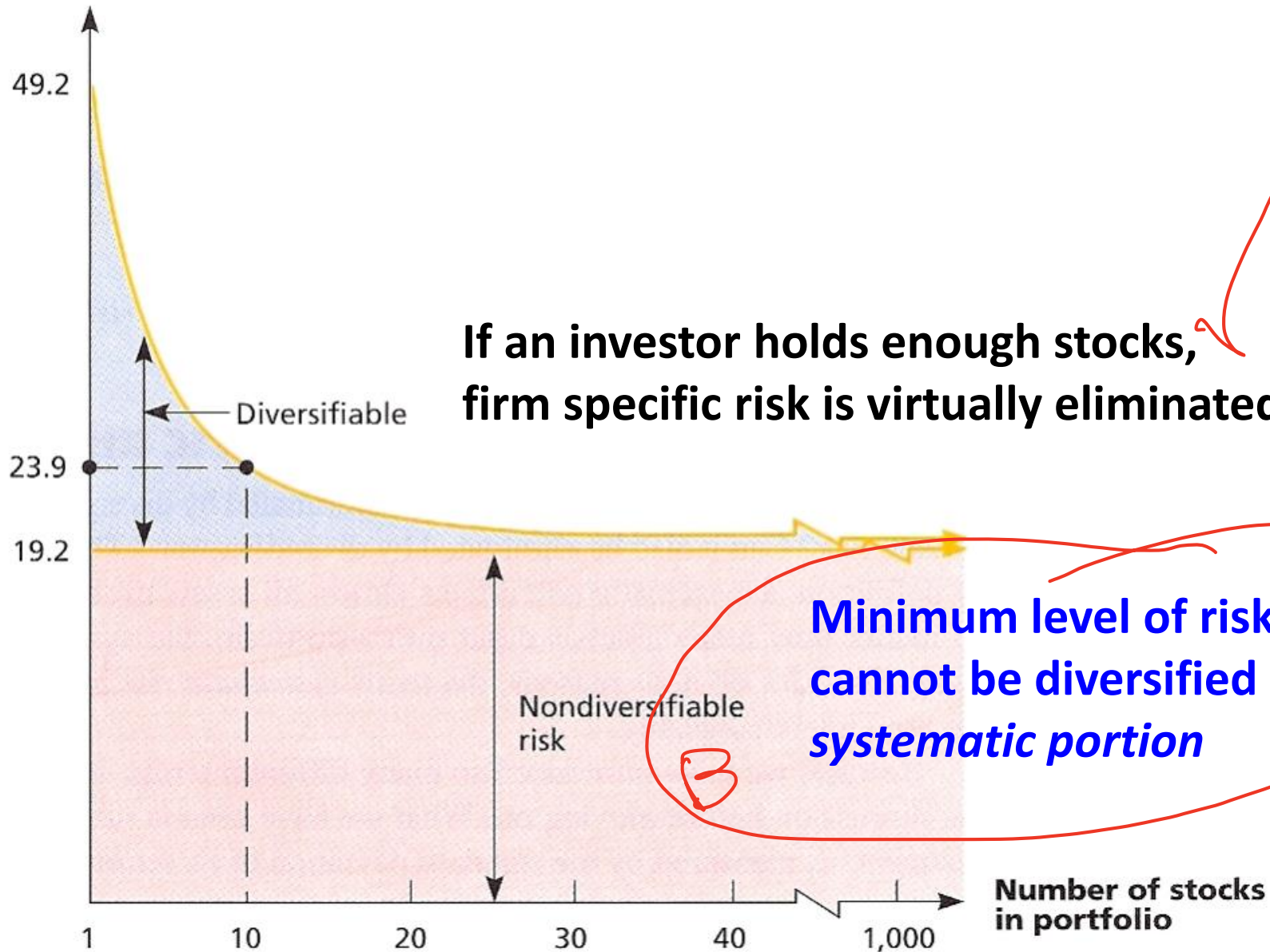
- *Risk due to overall market conditions*
- *Factors that affect a large number of assets*
- *Examples: changes in GDP, inflation, interest rates, etc.*
- ***Diversification does not reduce market related (systematic) risk***

Unsystematic Risk (Firm Specific Risk)

- *Risk factors that affect a limited number of assets*
- *Risk due to factors within the firm*
- *Examples: labor strike, parts shortages, major government contract is lost unexpectedly.*
- ***Diversification can effectively eliminate firm specific (unsystematic) risk***

Diversification

Average annual
standard deviation (%)



Risk and Rates of Return

Firm Specific risk - we can diversify away

– *it is mainly the market related risk we are concerned about as an investor*

Market risk –

we need to measure the sensitivity of the individual company's stock returns to the variability of returns of the market

– *How do we do this?*

Beta β

What is Beta? #1) STOCK RETURNS

- Beta measures a stock's volatility - *the degree to which its return fluctuates in relation to the overall market*

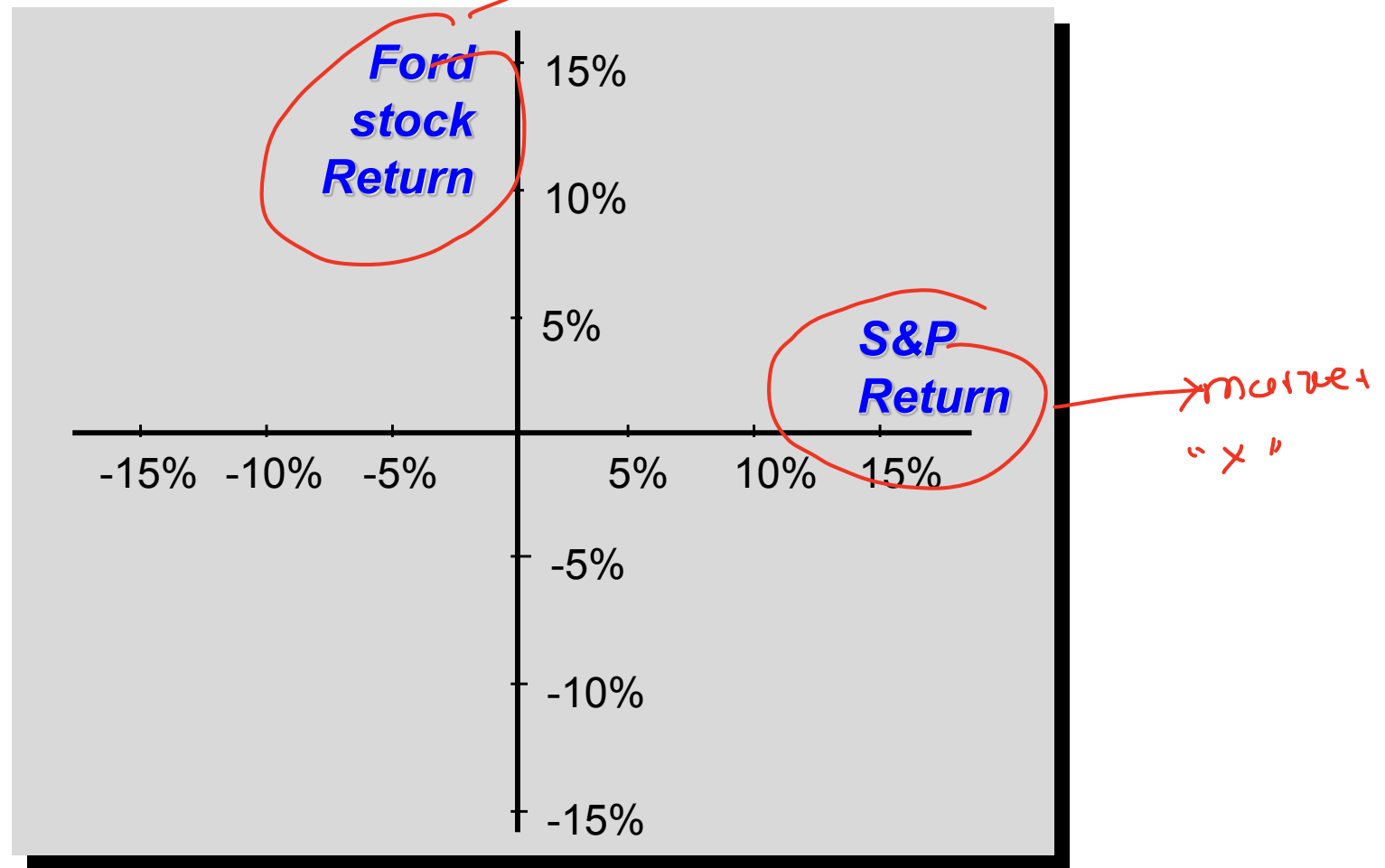
(We will use the S&P 500 as a proxy for the market)

How do we calculate Beta? #2) MARKET RETURNS

- Plot individual stock returns *vs.* returns of the market
- *Beta is the slope* of the regression line

Beta Calculation

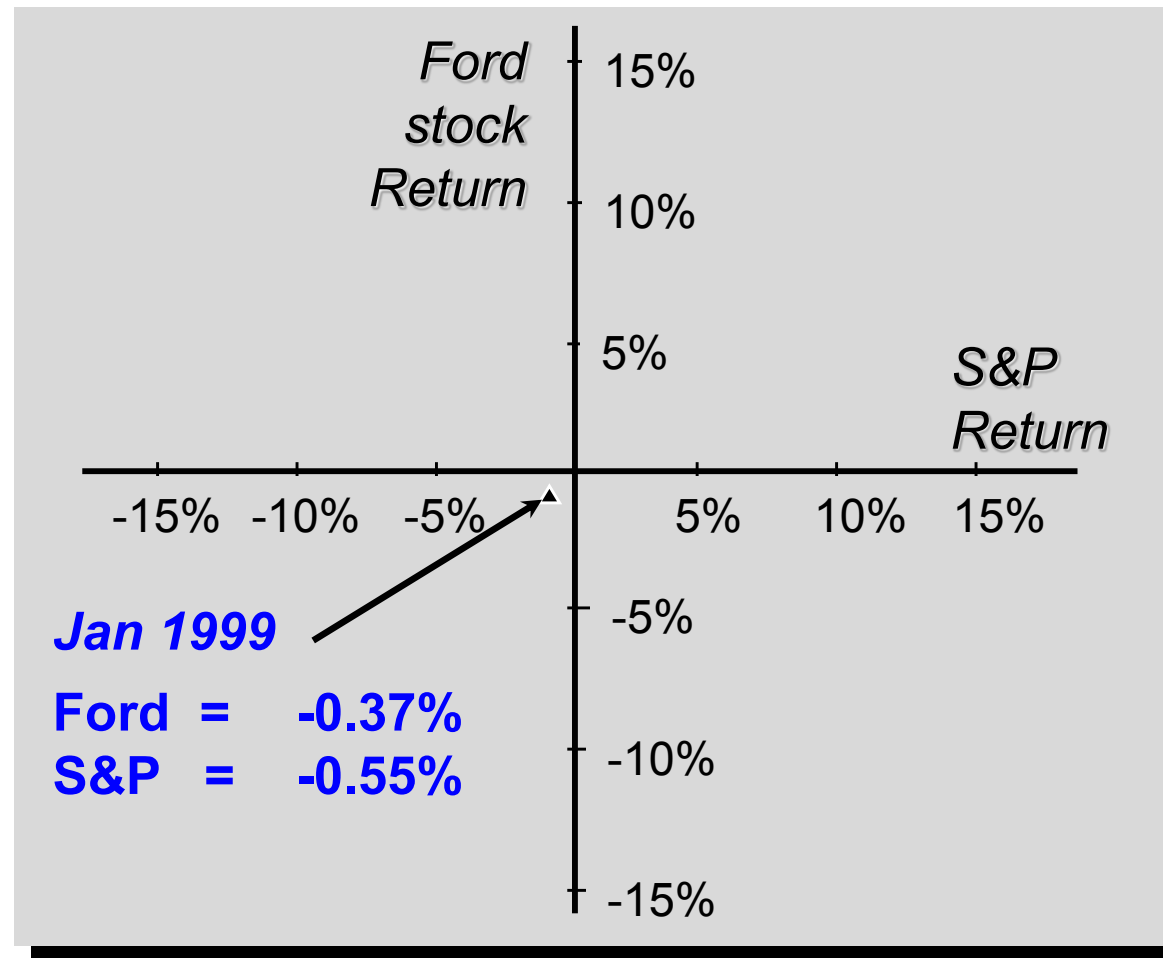
- Plot individual stock returns (Ford) vs. Market index (S&P 500) ^{↑ S&P}



Beta Calculation

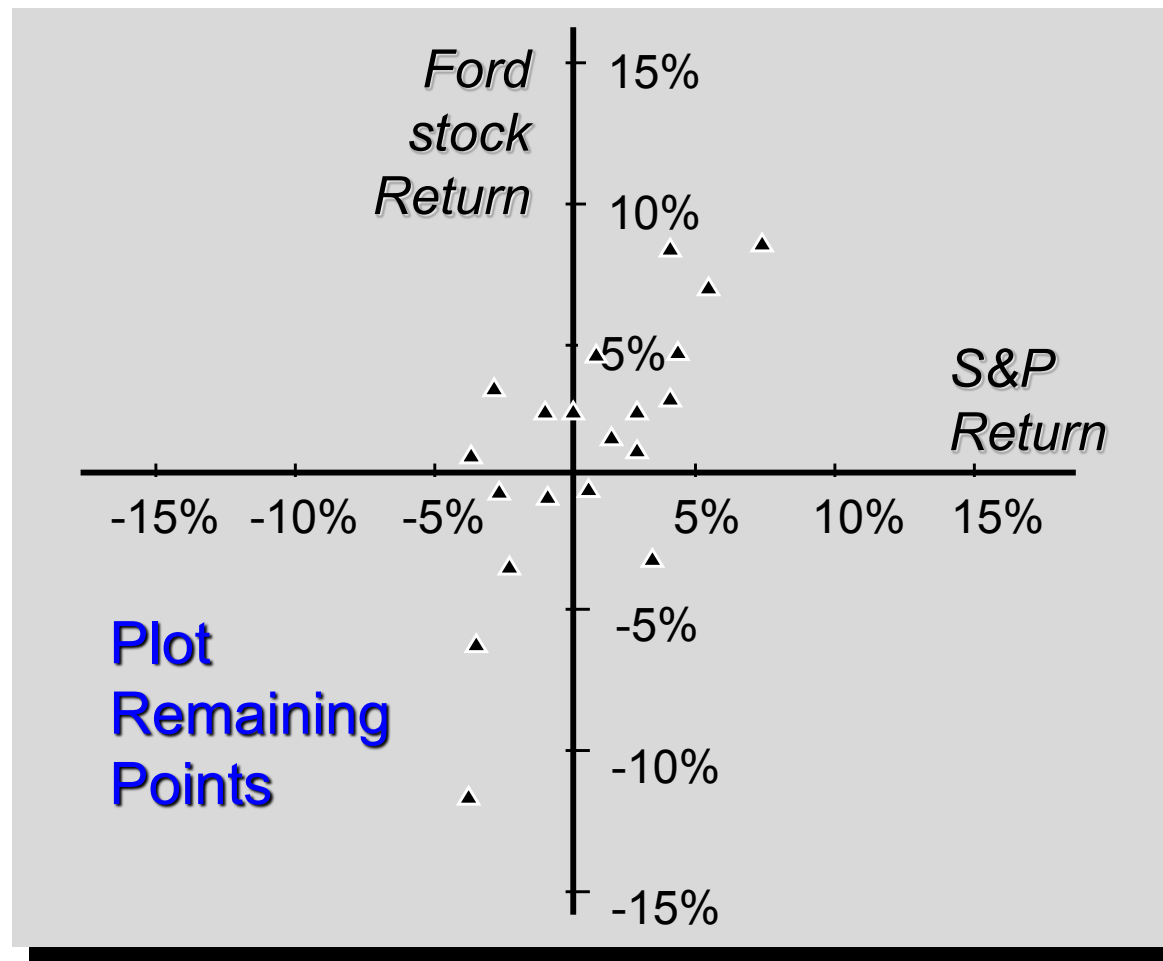
- Plot individual stock returns (Ford) vs. Market index (S&P 500)

$$\text{Beta} = \frac{\text{rise}}{\text{run}}$$



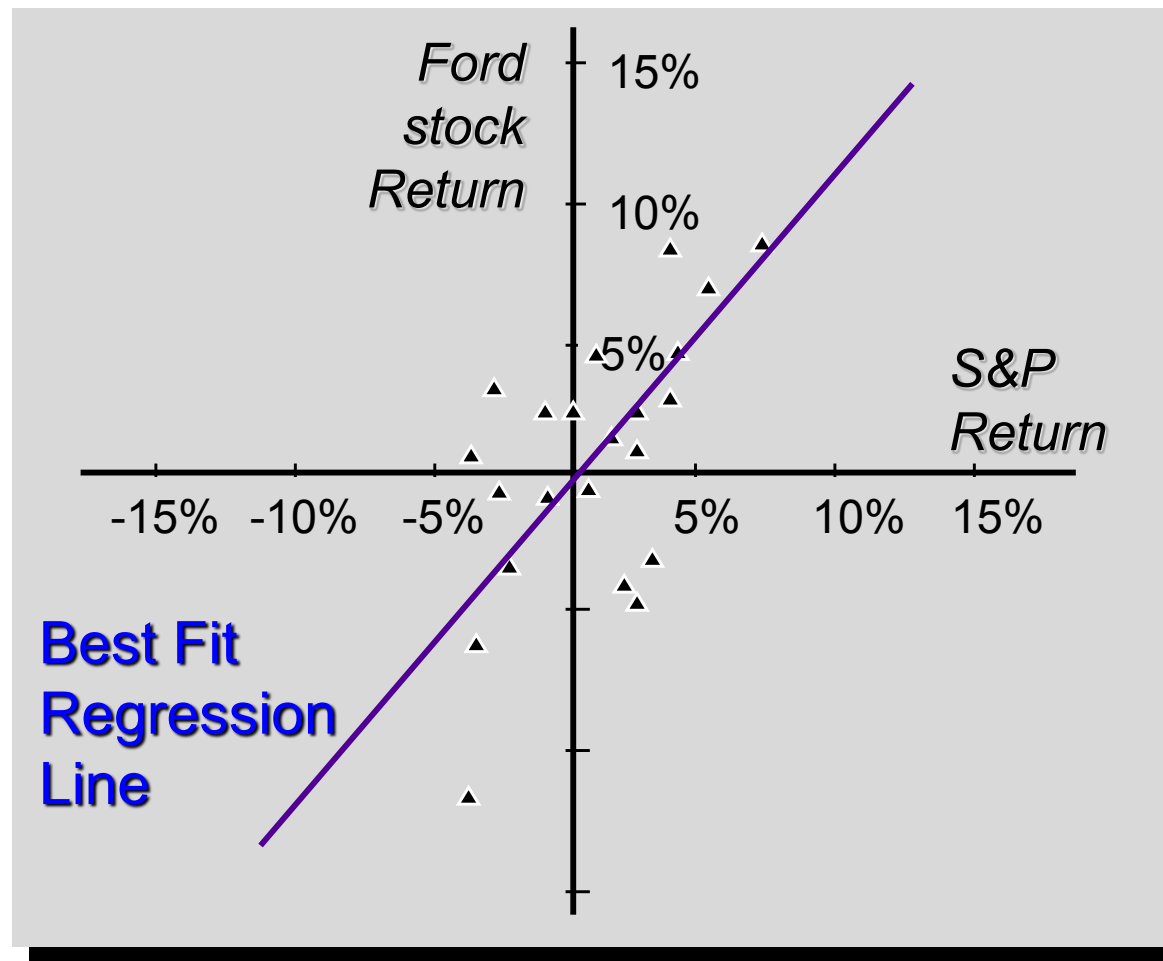
Beta Calculation

- Plot individual stock returns (Ford) vs. Market index (S&P 500)



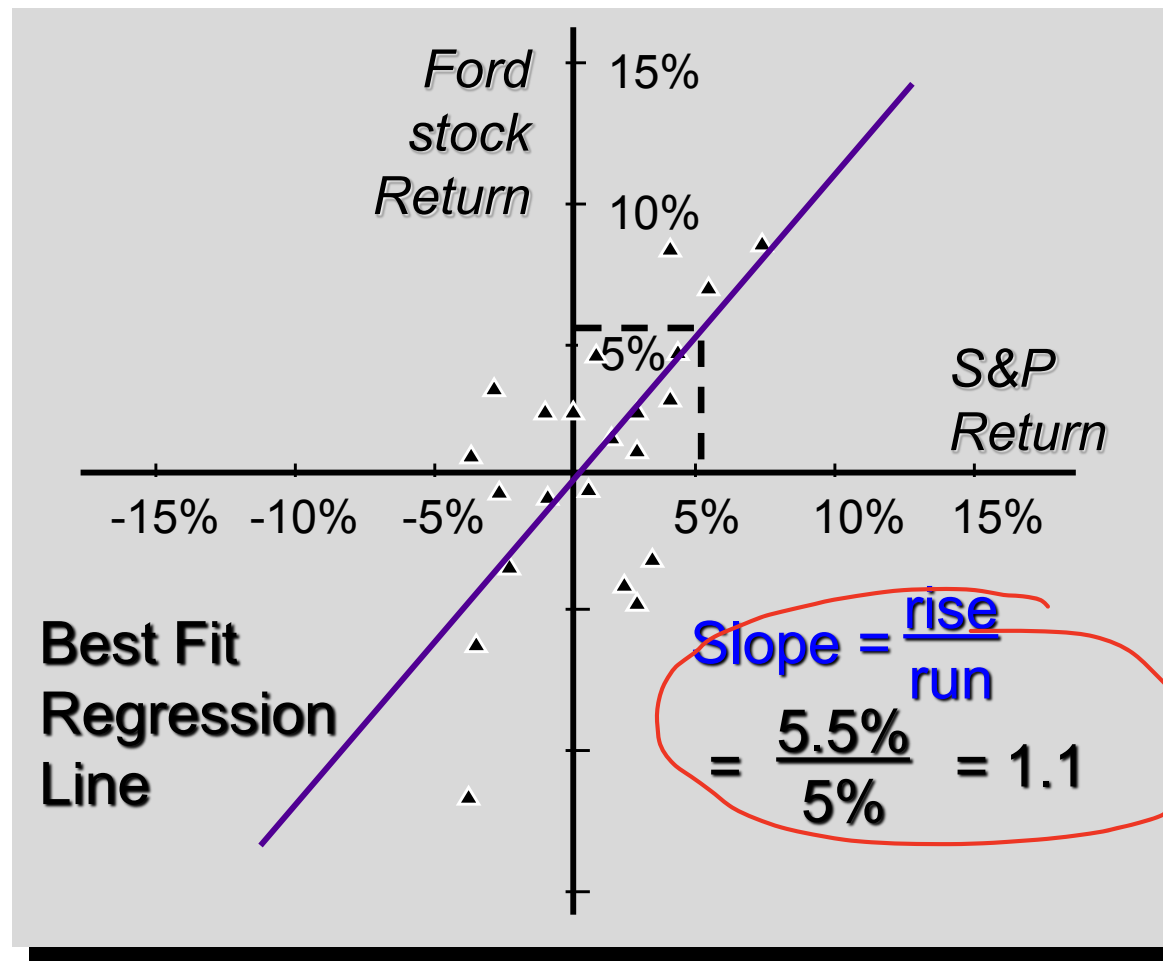
Beta Calculation

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Beta Calculation

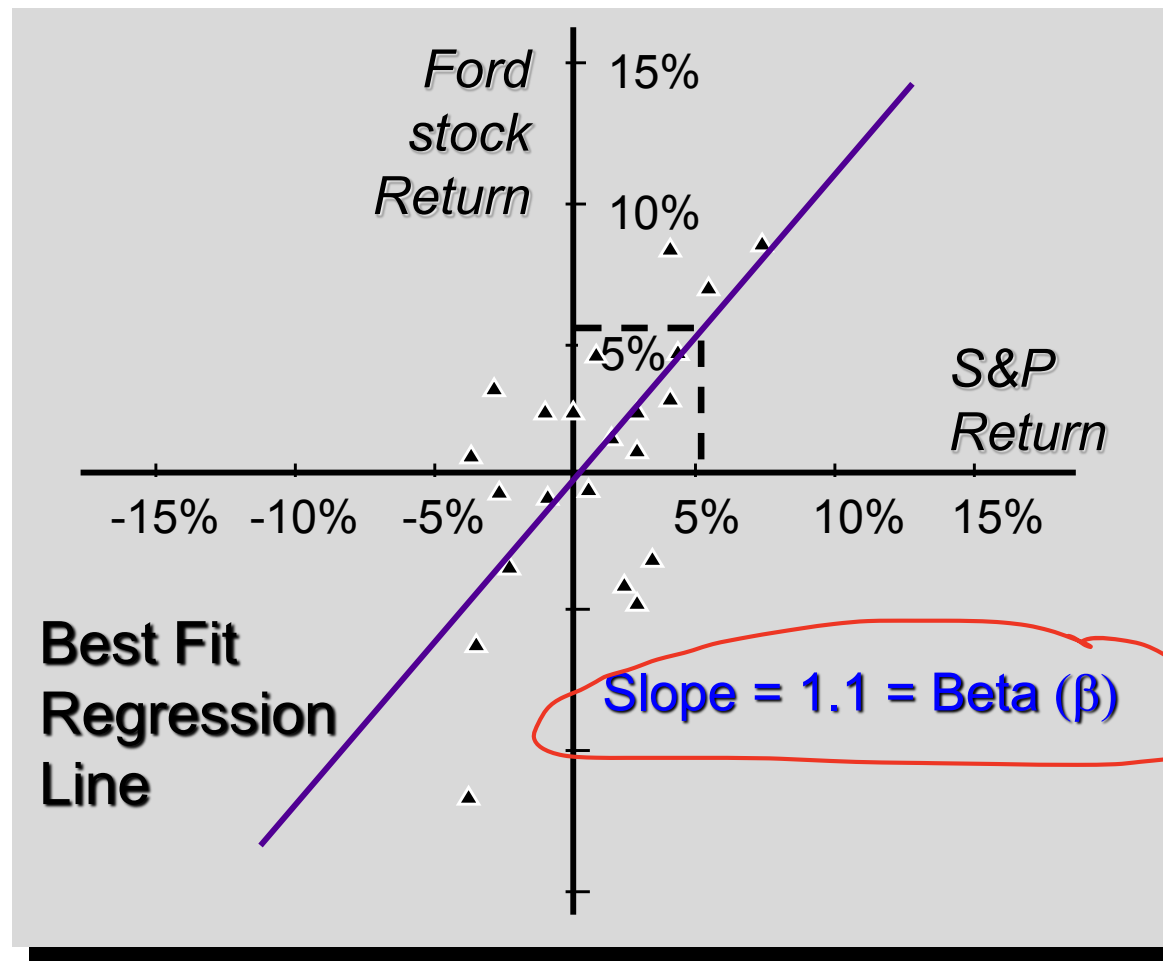
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Degree to which stock returns fluctuates relative to market...

Beta Calculation

- Plot individual stock returns (Ford) vs. Market index (S&P 500)



Degree to which stock returns fluctuates relative to market...

Beta calculation

$\beta_{rf} = ?$

$$\beta = \frac{\text{Cov}(r_i, r_m)}{\text{Var}(r_m)}$$

$\beta_{\text{market}} =$
 $r_i = \text{stock}$
 $r_m = \text{market}$

$\beta = \text{slope (known } y' \text{'s, known } x' \text{'s)}$

$\beta = \text{linest} ()$

$\beta = x \text{ coefficient from linear regression}$

Beta calculation

Open “week 15 – BETA” file

calculate the beta of TSLA and EXEL using the 3 approaches below:

$$\beta = \frac{Cov(r_i, r_m)}{Var(r_m)}$$

$\beta = \text{slope } (y's - \text{stock}, x's - \text{market})$

$\beta = \text{x coefficient from linear regression}$

Portfolio Beta

The portfolio beta is the weighed average of the individual betas:

$$\beta_p = (W_a * \beta_a) + (W_b * \beta_b)$$

Using Beta β

Beta = 1

has volatility similar to the market

Beta < 1

Lower Risk/volatility Company

Return on stock will be less volatile than the market on average

Example: Utility Companies

Beta > 1

Higher Risk/volatility Company

Stock return will be more volatile than the market on average

Example: Technology Companies

Capital Asset Pricing Model (CAPM)

Definition:

The expected return of any security equals the default free, risk free rate plus the market risk premium modified by how the security return varies with the market's return (beta).

Capital Asset Pricing Model (CAPM)

The **CAPM** measures *expected rate of return* for investments, given the degree of market risk measured by beta:

$$E(R_i) = R_f + [E(R_m) - R_f]\beta$$

where:

$E(R_i)$ = expected return on investment

R_f = risk-free rate of return

$E(R_m)$ = expected return on the market

β = Beta

Factors Affecting Required Return

$$E(R_i) = R_f + [E(R_m) - R_f]\beta$$

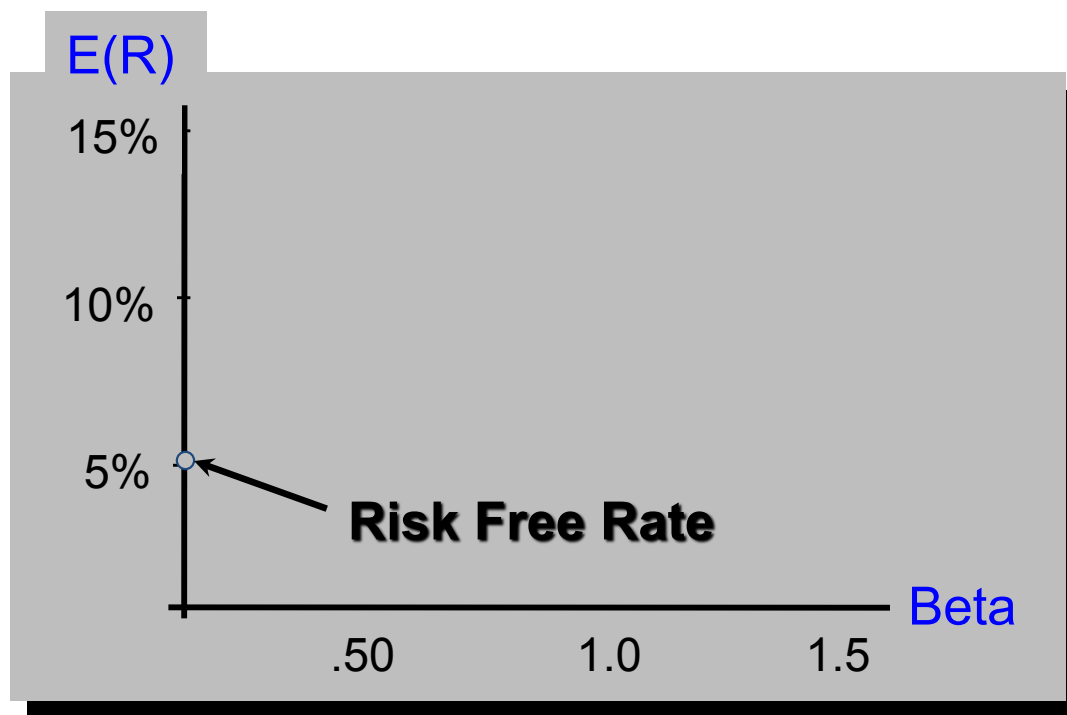
- R_f measures the pure time value of money
- $[E(R_m) - R_f]$ measures the reward for bearing systematic risk
- β measures the amount of systematic risk

Security Market Line (SML)

Suppose that:

return on the market $E(R_m) = 12\%$

risk free rate $R_f = 5\%$



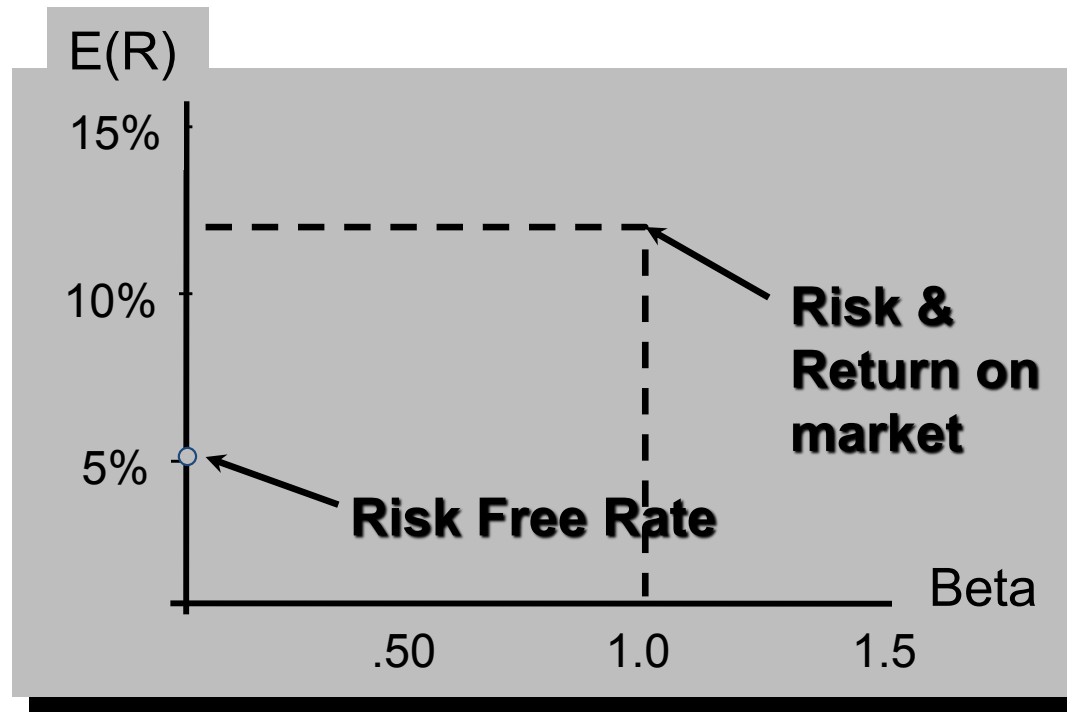
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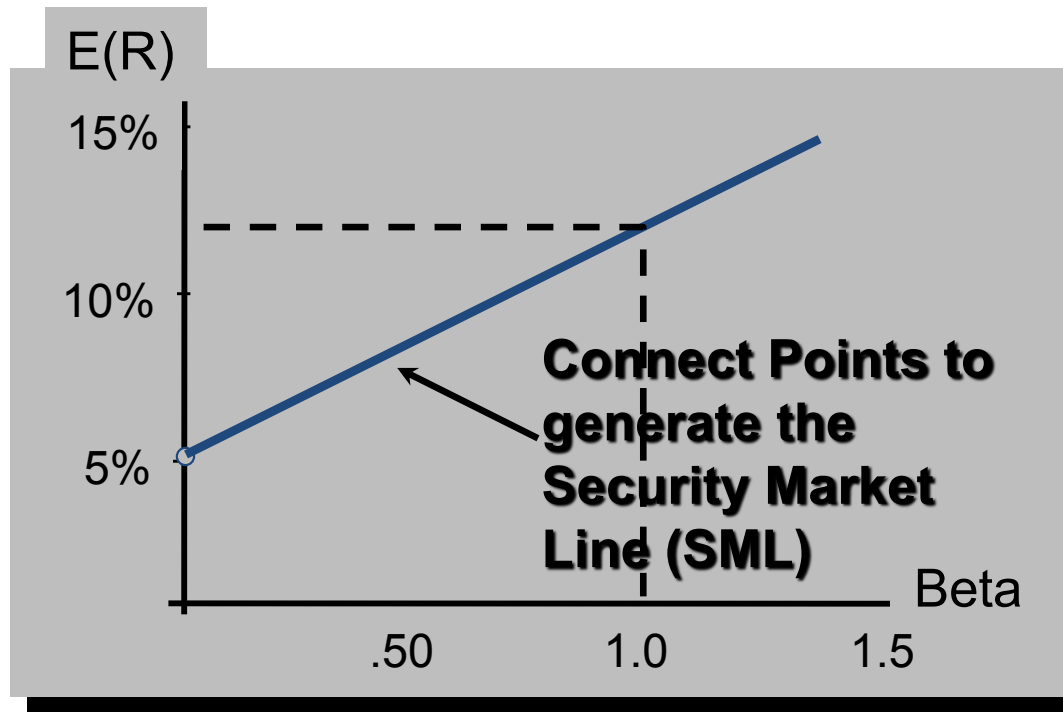
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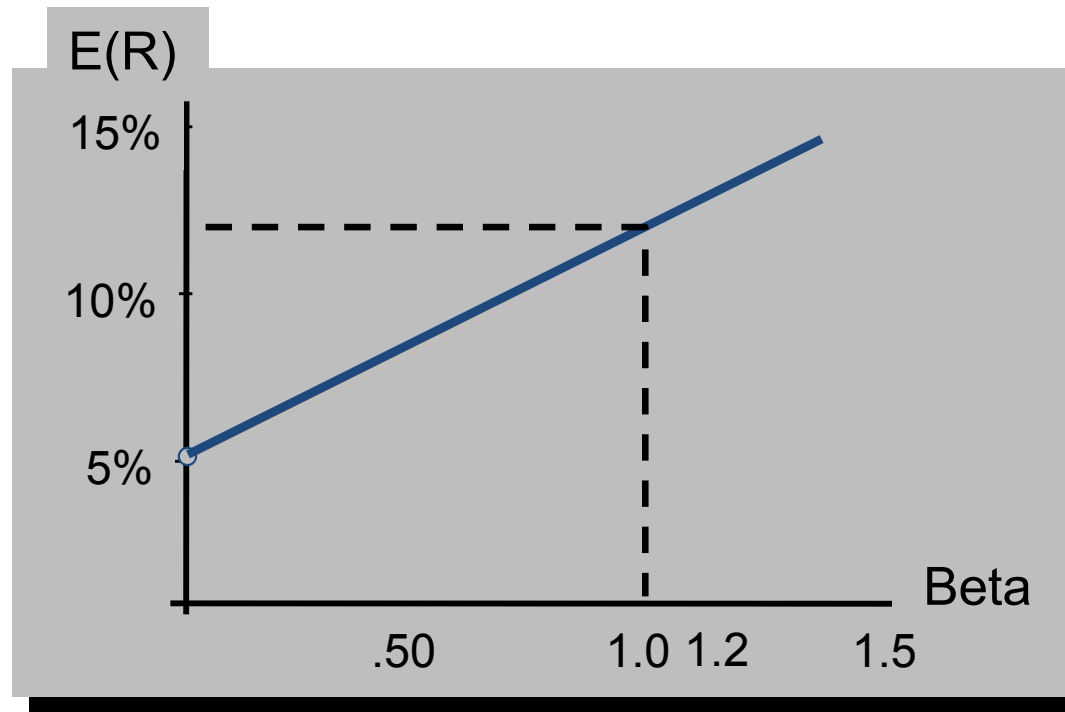
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Security Market Line (SML)

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**Given a beta,
We can
calculate
expected
return**

$$E(R_i) = R_f + [E(R_m) - R_f]\beta$$

